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ADMS

16.25.1

DNOM2HDB Installation Guide

Copyright

Change History

Date	Change Description
Jan 26, 2023	ADMS 3.16: • Editorial and formatting updates.
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1. Overview

This chapter provides an overview of the installation process and describes the conditions that must be met prior to installing DNOM2HDB.

1.1. Software Organization

The software that is delivered with DNOM2HDB consists of:

- Habitat codeset: DNOM2HDB

1.2. Prerequisites

DNOM2HDB can be installed on Habitat , Platform (or Scada), and ADMS for Microsoft Windows and Red Hat Enterprise Linux operating systems.

1.3. Installation Process

1. Install the DNOM2HDB files using the `ADMS_dnom2hdb_X64.msi` kit.
2. Manually build the DNOM2HDB files.

2. Software Installation

This chapter describes the DNOM2HDB installation process.

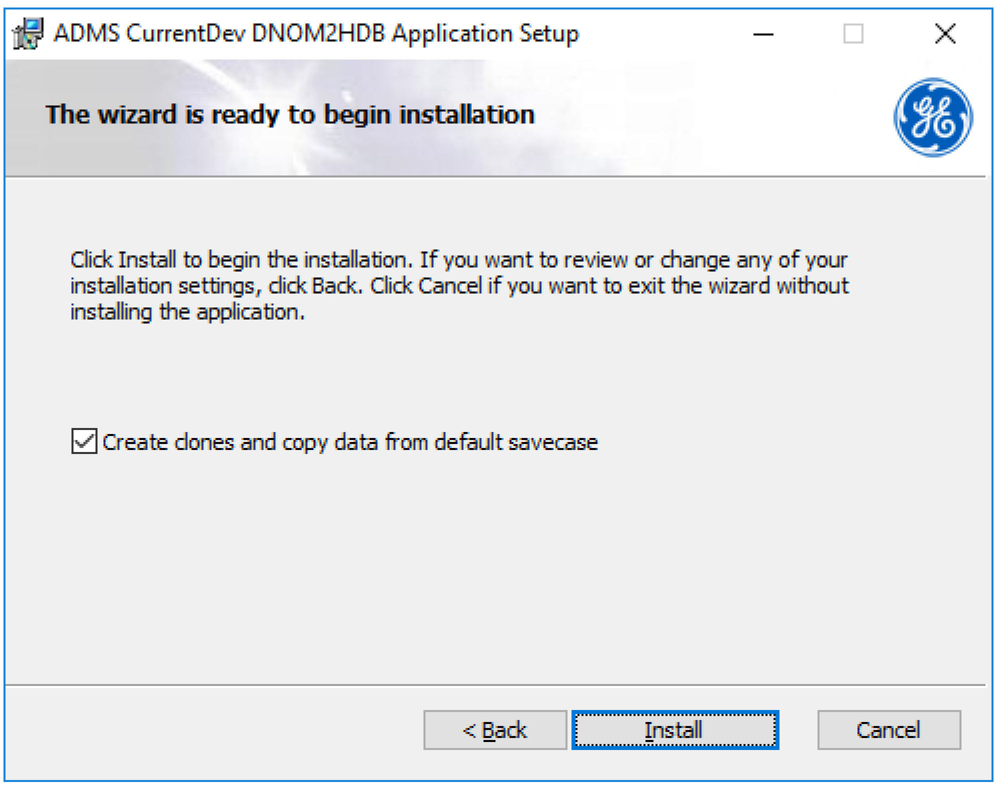
2.1. Installing DNOM2HDB

This section describes how to install the DNOM2HDB files using the `ADMS_dnom2hdb_X64.msi` kit.

> **Caution**
>
> The DNOM2HDB installation stops all Habitat applications that are running on the computer.

To install the DNOM2HDB files:

1. Run the ADMS_dnom2hdb_X64.msi installer.
2. Select the Create clones and copy data from default savecase option and then click Install.



3. Wait until the installation completes and then click Finish.

During the installation, DNOM2HDB files are copied to the following directories:

- D:\Eterra\habsrc06\habuser\special\Dnom2hdb\
- D:\Eterra\habweb06\displays\dnom2hdb\
- D:\Eterra\habdata06\appdefs\

- D:\Eterra\habdata06\cdbroot\clones\
- D:\Eterra\habdata06\cdbroot\dictionary
- D:\Eterra\habdata06\savecases\
- D:\Eterra\habdata06\uidbroot\ddls\
- D:\Eterra\habitat06\habuser\bin\

2.2. Building DNOM2HDB Individually

This section describes how to build a DNOM2HDB application individually (that is, outside of a system build).

> **Note**
>
> Do not perform the procedure below if you installed DNOM2HDB using the `ADMS_dnom2hdb_X64.msi` installer.

2.2.1. Loadhab

At the command prompt, go to the DNOM2HDB folder (by default, `D:\Eterra \habsrc06 \habuser \special \Dnom2hdb`) and load the Habitat files using the command:

```
loadhab *
```

2.2.2. Clone

At the command prompt, go to the DNOM2HDB folder (by default, `D:\Eterra\habsrc06\habuser\special\Dnom2hdb`).

- For a standalone system (for example, operator simulator or study environment), create the clone using the command:

  ```
  hdbcloner -c create_clone -a _dnom2hdb_ -f _fam_
  ```

- For a dual redundant system (MRS is used), create the clone using the command:

  ```
  hdbcloner -c create_clone -a _dnom2hdb_ -f _ems_ -replicate
  ```

> **Note**
>
> The command "hdbcloner -replace" allows you to update a clone structure and keep the current data.

2.2.3. Build

Build the DNOM2HDB codeset, and verify that a `dnom2hdb.exe` file is created in the habitatxx/habuser/bin folder with a date that corresponds to the time that the BUILD command was issued.

> **Note**
>
> If the Windows build fails because it cannot find some header files, verify that `habitat_environment_habdev_net_template.bat` sets up proper INCLUDE paths for the SDK kit. It is possible to manually set paths for VS2008: Tools/Projects/Projects and Solutions/VC++ Directories.

For more information, refer to the *ESCATOOLS User Guide*.

2.3. Validating/Verifying the DNOM2HDB Installation

Verify that the `dnom2hdb.exe` application can be launched from the Windows or Linux command prompt.

At the Windows command prompt, type:

```
> context dnom2hdb ems
> dnom2hdb
```

At the Linux command prompt, type:

```
% context dnom2hdb ems
% $HABUSER_BINDIR\dnom2hdb
```

> **Note**
>
> Habitat must be running to start `dnom2hdb.exe`.

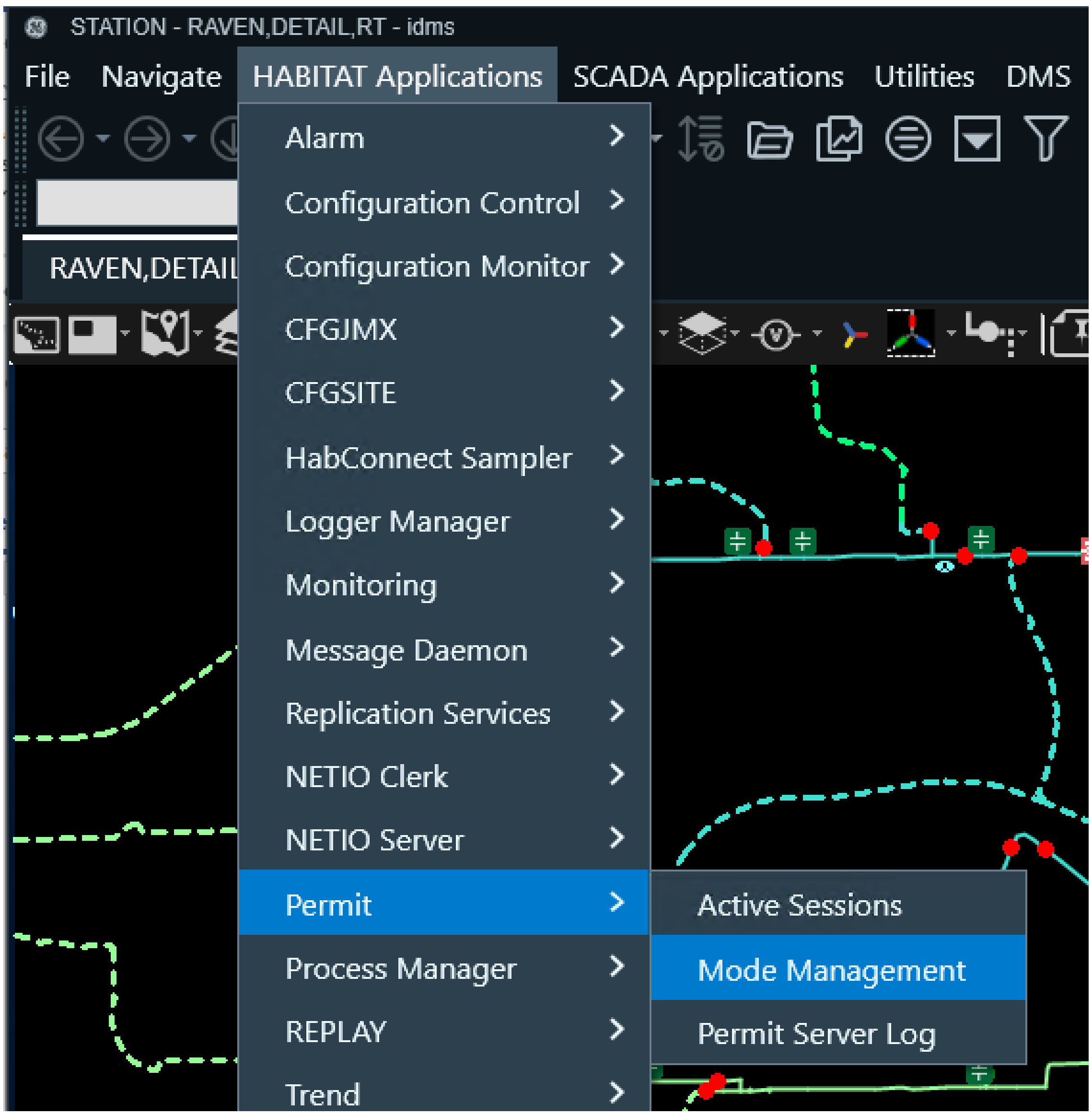
3. System Configuration

This chapter discusses some specific configurations that are required after a DNOM2HDB installation. For more information about any required steps, refer to the Habitat documentation.

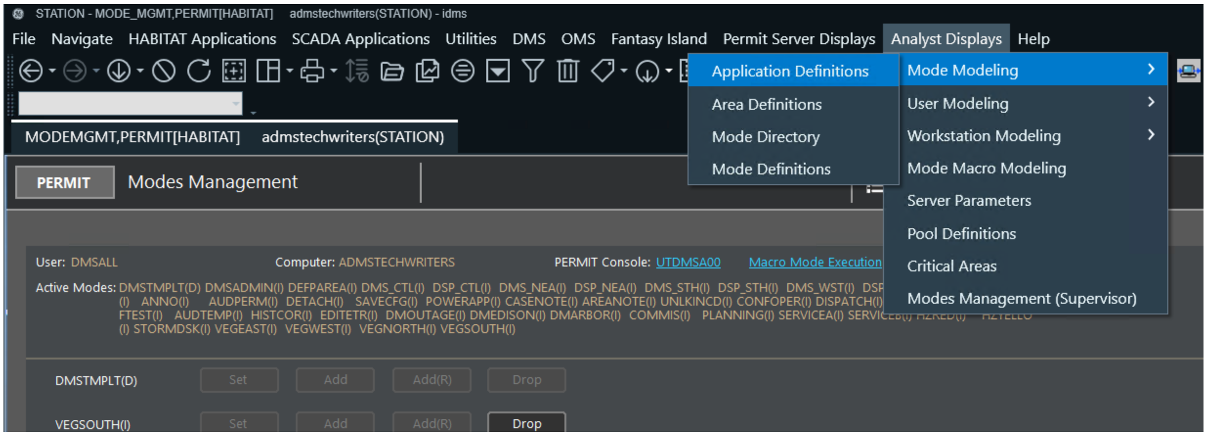
3.1. PERMIT

3.1.1. Defining the DNOM2HDB Application

1. In the Browser client, click HABITAT Applications > Permit > Mode Management.

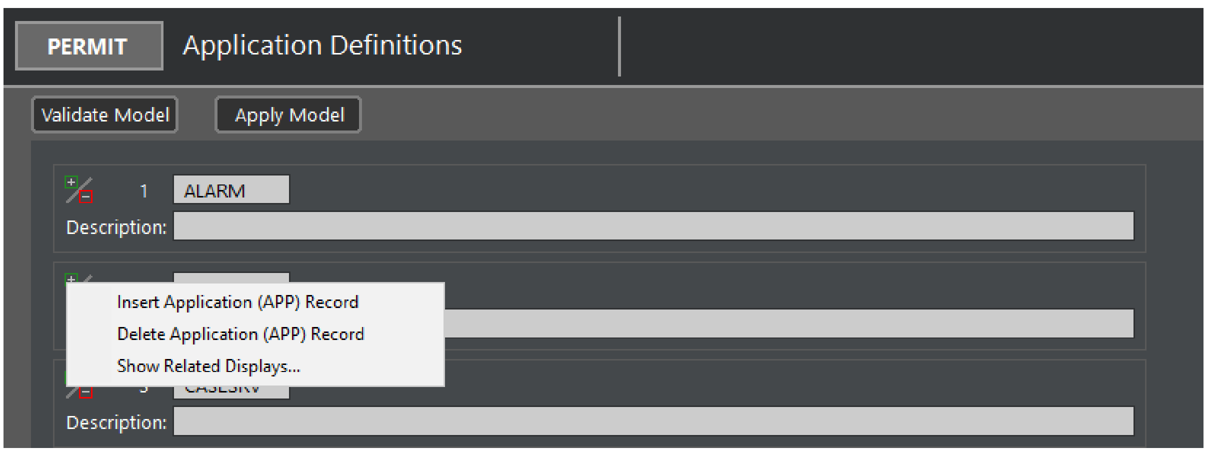


2. On the Analyst Displays menu, click Mode Modeling > Application Definitions.



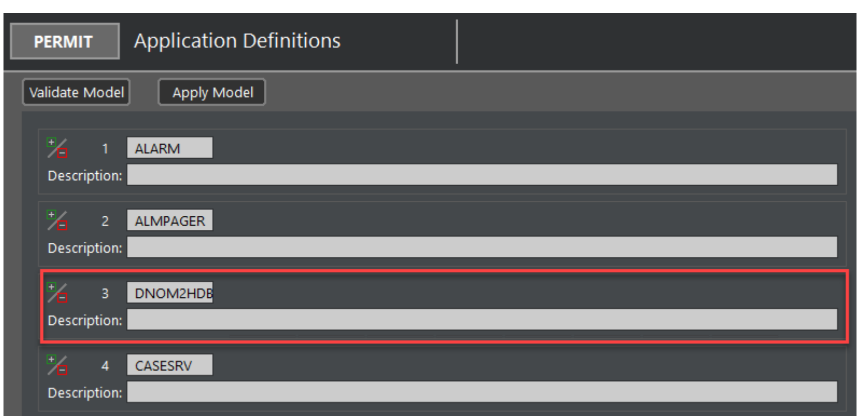
This opens the Application Definitions display.

3. On the Application Definitions display, right-click the Insert or Delete Application Record field and select Insert Application (APP) Record.



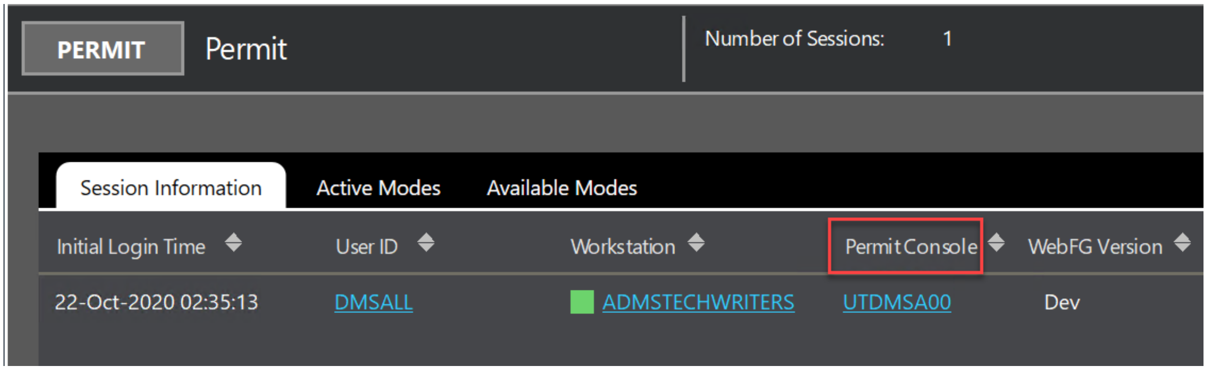
A new application record is added.

4. Enter the DNOM2HDB application name and press Enter.

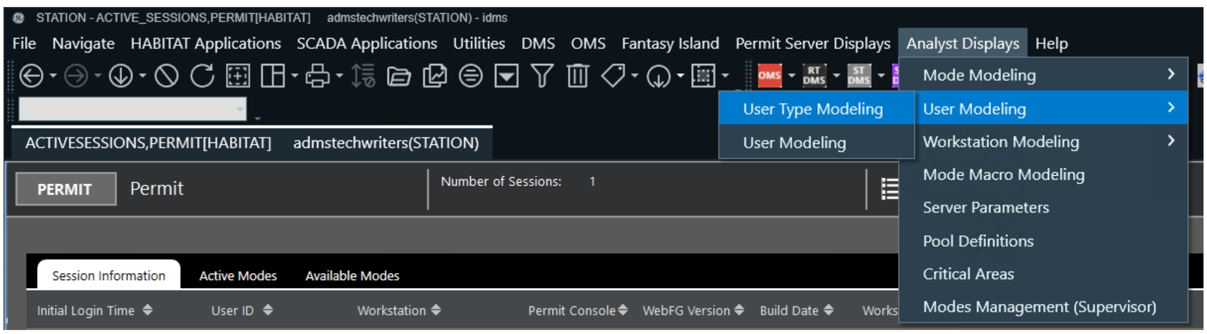


5. Open the Permit display (HABITAT Applications menu > Permit > Active Sessions).
6. On the Permit display, identify your user type on the Permit Console.

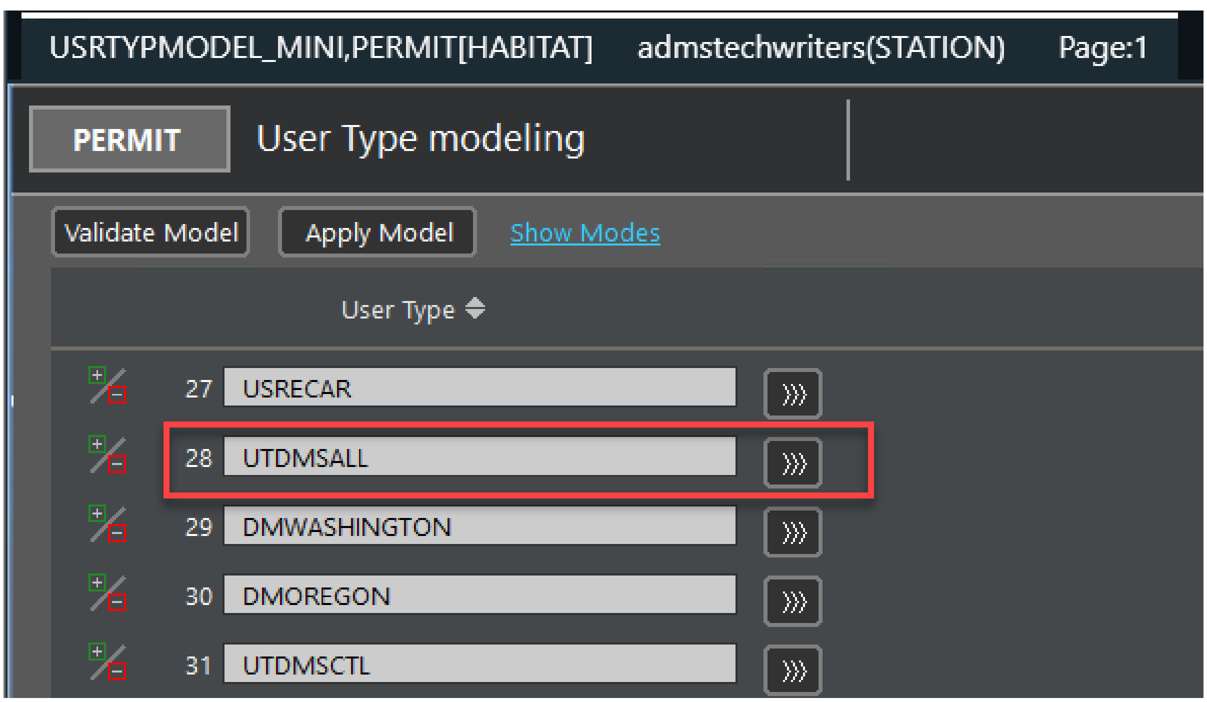
In this example, the user type is UTDMSA00.



7. Go to the User Type Modeling analyst display (Analyst Displays menu > User Modeling > User Type Modeling).

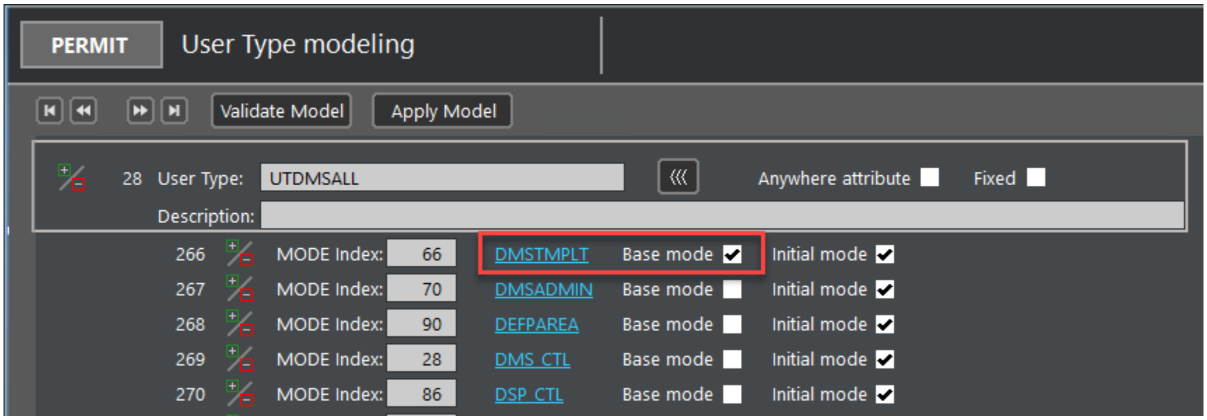


8. On the User Type Modeling display, click the expand (**>>>**) button for the identified user type (in this case, UTDMSALL).



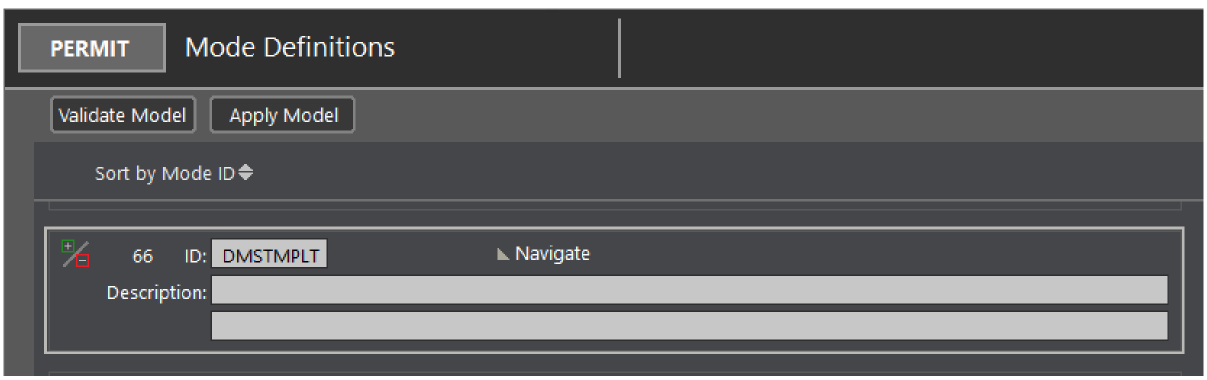
9. Identify the base mode for the user type.

   In this example, the base mode is DMSTMPLT.

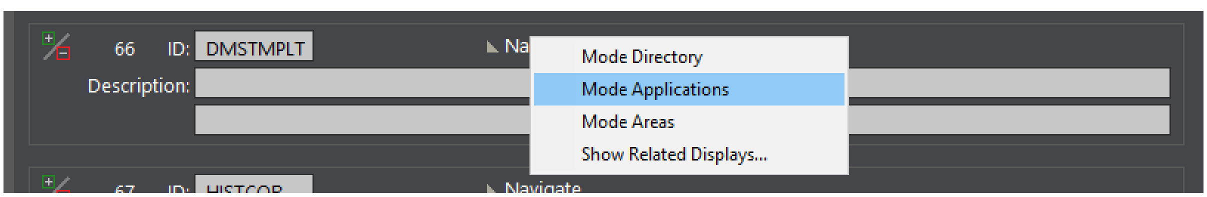


10. Click the Base Mode name.

    The Mode Definitions display is shown for the base mode.

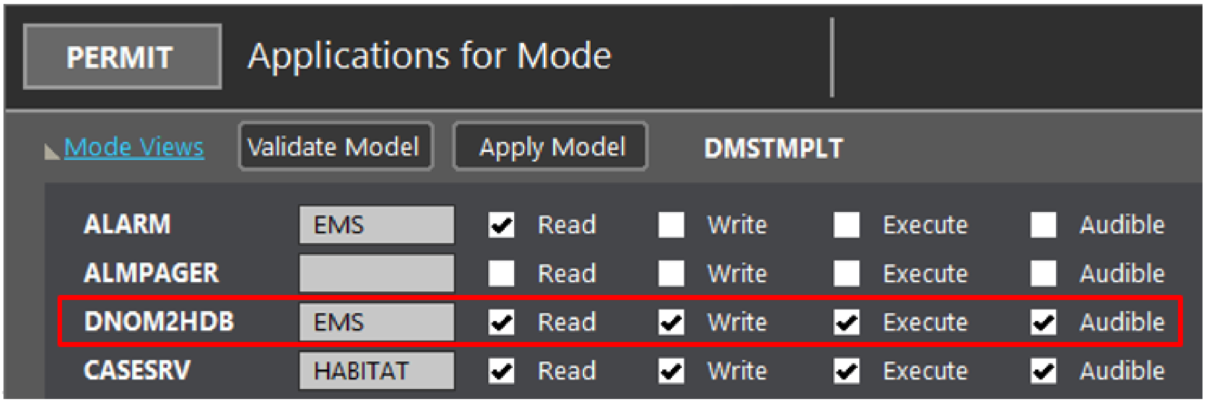


11. Right-click the Navigate field and select Mode Applications.

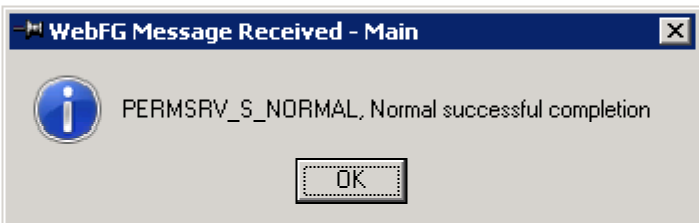


    The Applications for Mode display is opened.

12. Scroll to where DNOM2HDB is added as a PERMIT application.
13. Enter the EMS family name, select the permission boxes, and then press Enter.



14. Click Validate Model and check that the model validated successfully.

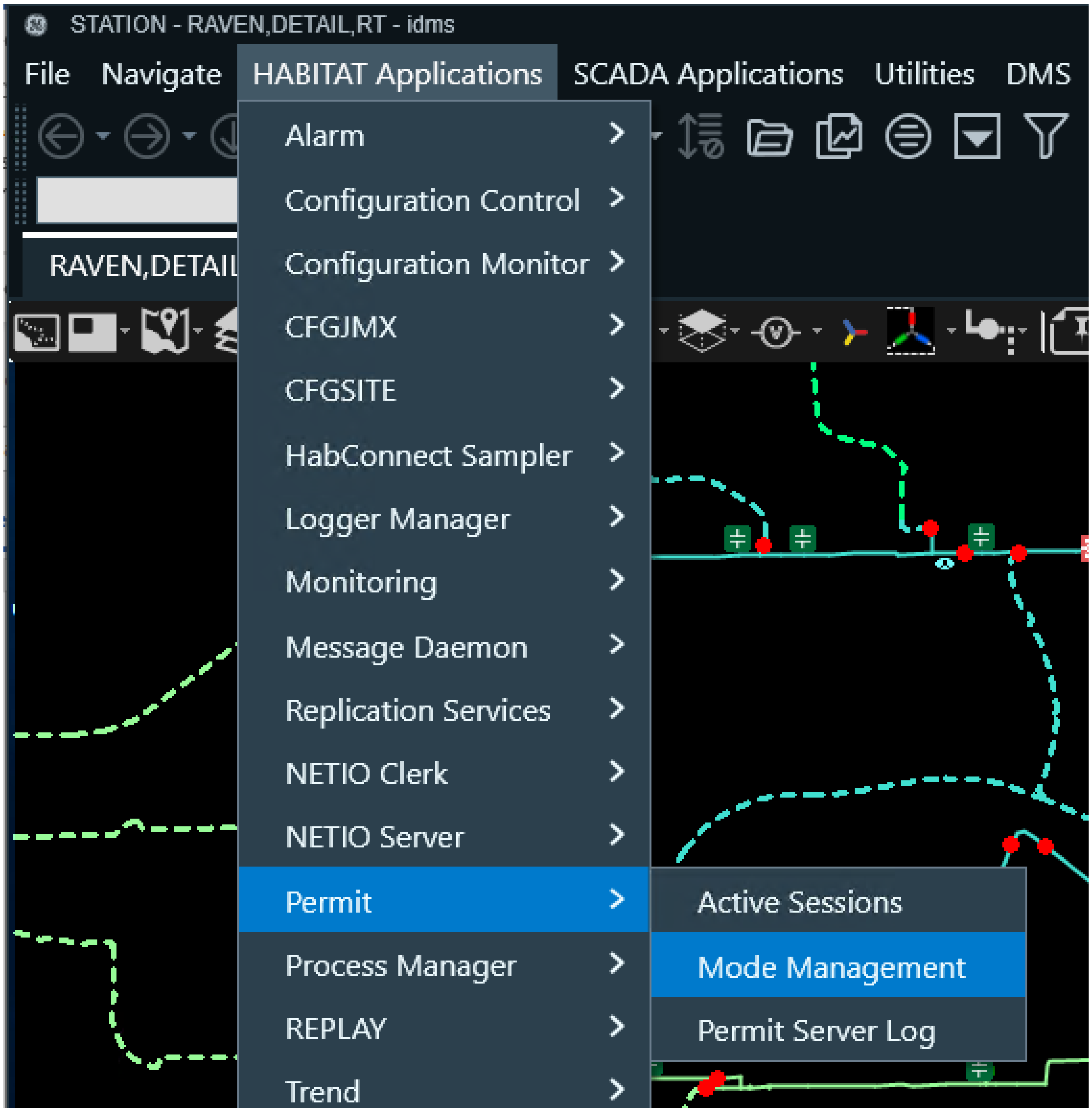


15. Click Apply Model and check that the model was applied successfully.

3.1.2. Defining the DNAFAPP Application

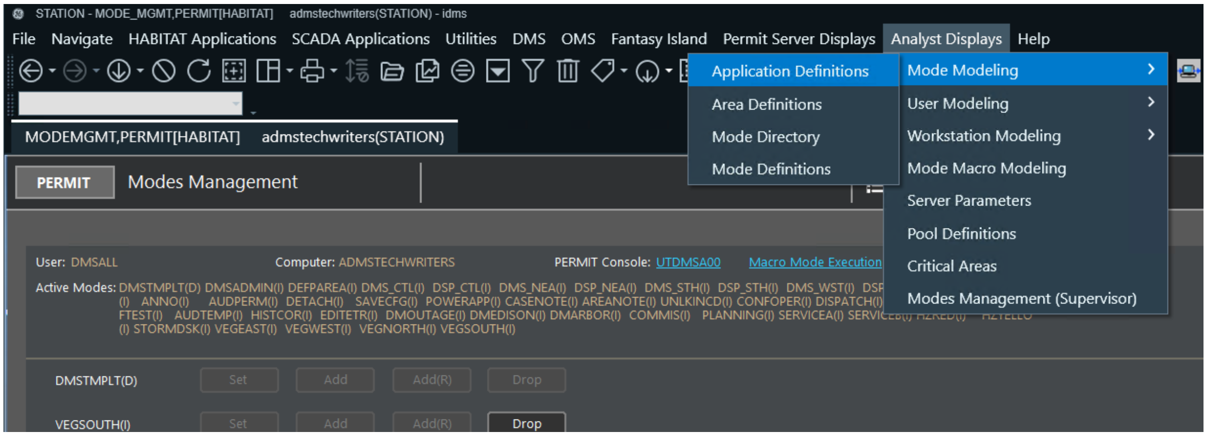
The DNAFAPP application is required to show DNAF-related DNOM2HDB displays, such as DNAF_LVMSTEP, DNAF_FISRSTEP, and DNAF_FISRPLAN.

1. In the Browser client, click HABITAT Applications menu > Permit > Mode Management.



The Mode Management display is opened for PERMIT.

2. On the Analyst Displays menu, click Mode Modeling > Application Definitions.

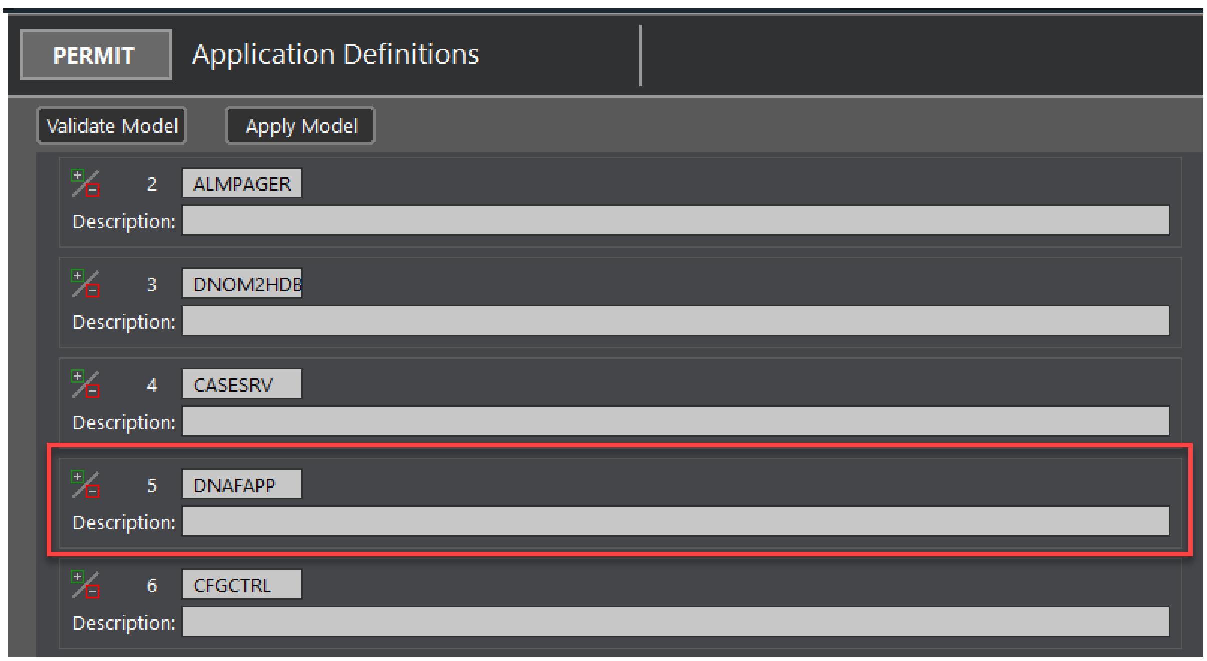


This opens the Application Definitions display.

3. On the Application Definitions display, right-click on the Insert or Delete Application Record field and select the Insert Application (APP) Record option.

   A new application record is added.

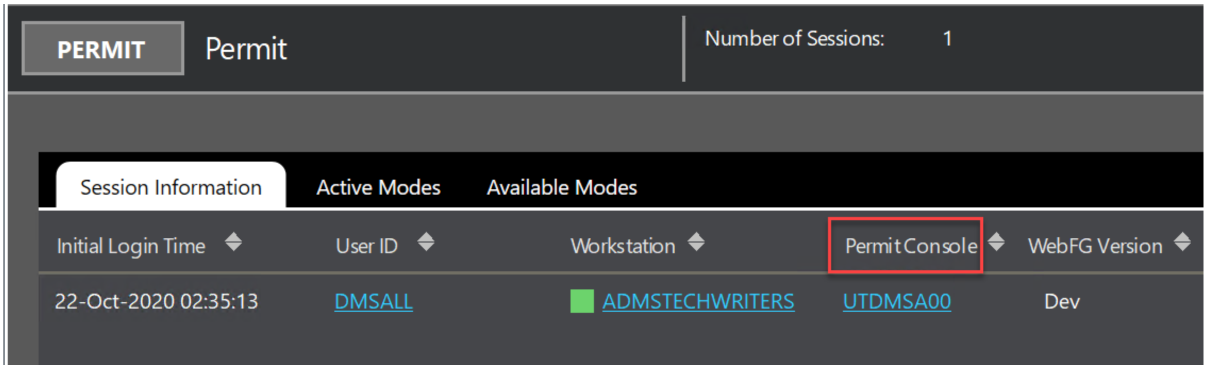
4. Enter the DNAFAPP application name and press Enter.



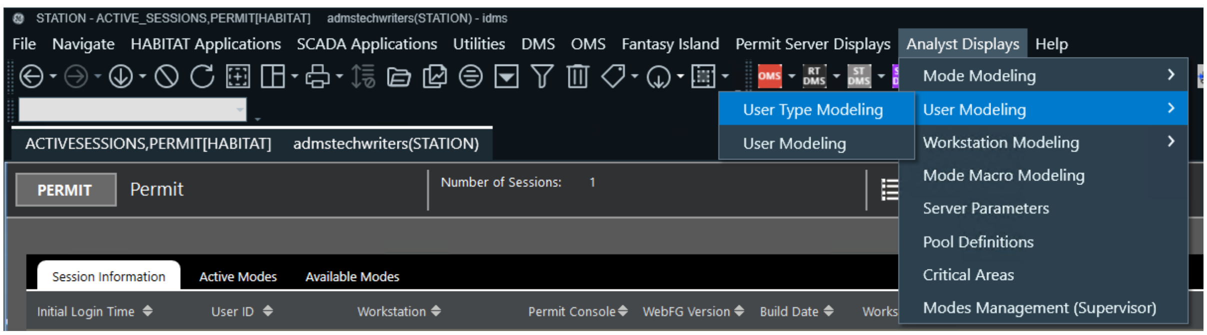
5. Open the Permit main display (HABITAT Applications menu > Permit > Active Sessions).

6. On the Permit main display, identify your user type on the Permit Console.

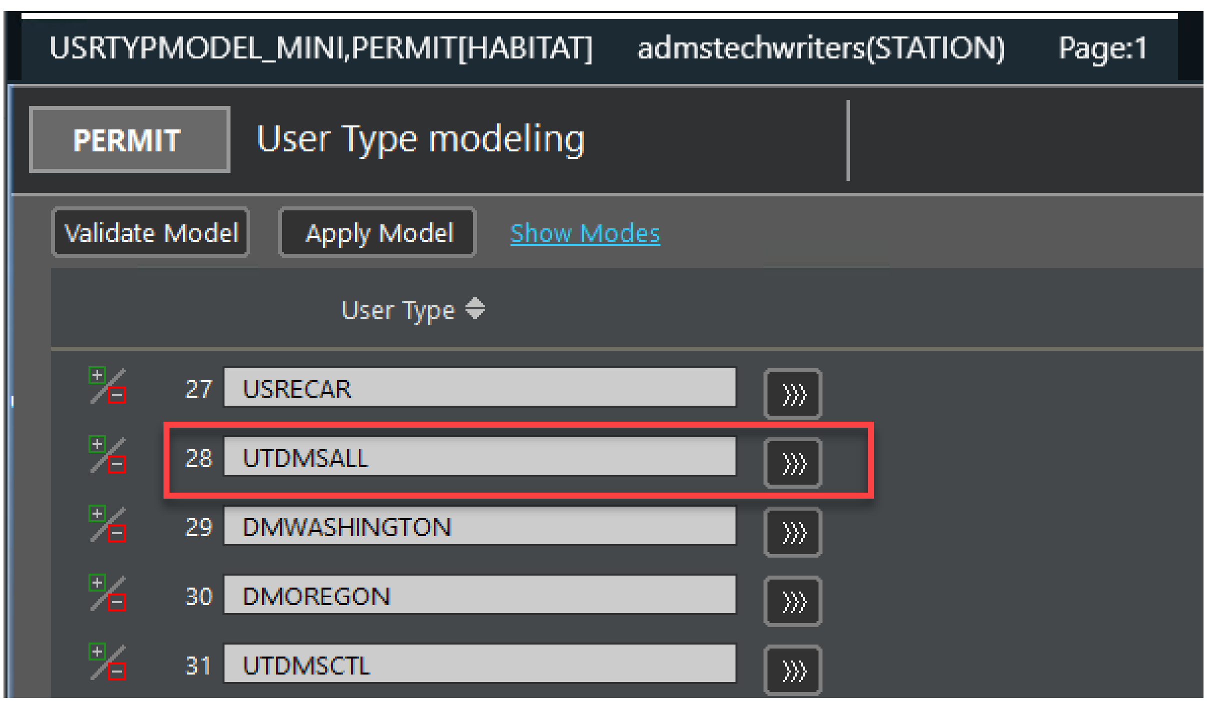
   In this example, the user type is UTDMSA00.



7. Go to the User Type Modeling analyst display (Analyst Displays menu > User Modeling > User Type Modeling).

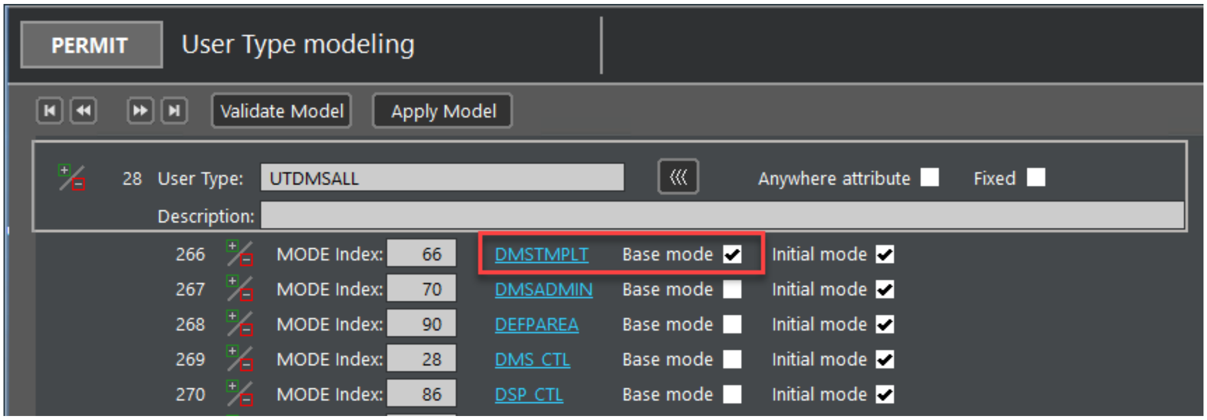


8. On the User Type Modeling display, click the expand (**>>>**) button for the identified user type (in this case, UTDMSALL).



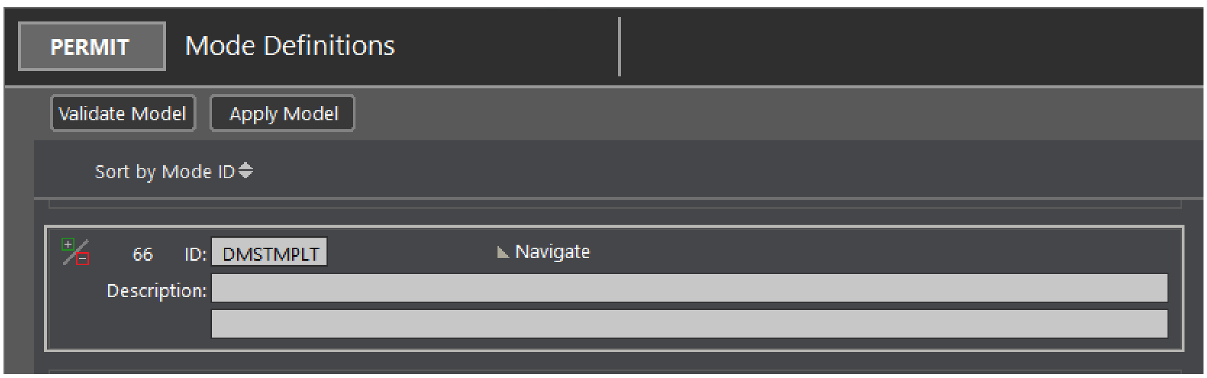
9. Identify the base mode for the user type.

In this example, the base mode is DMSTMPLT.

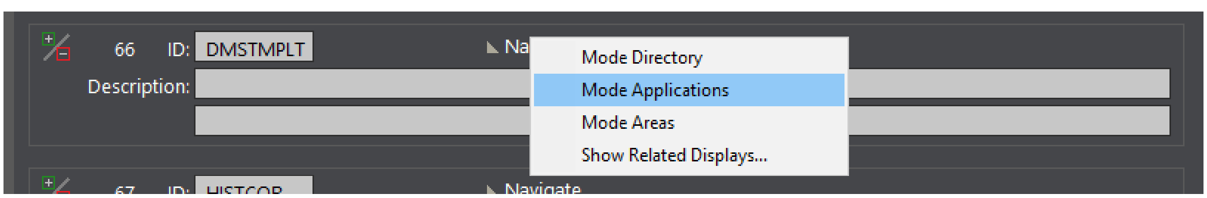


10. Click the Base Mode name.

    The Mode Definitions display is shown for the base mode.

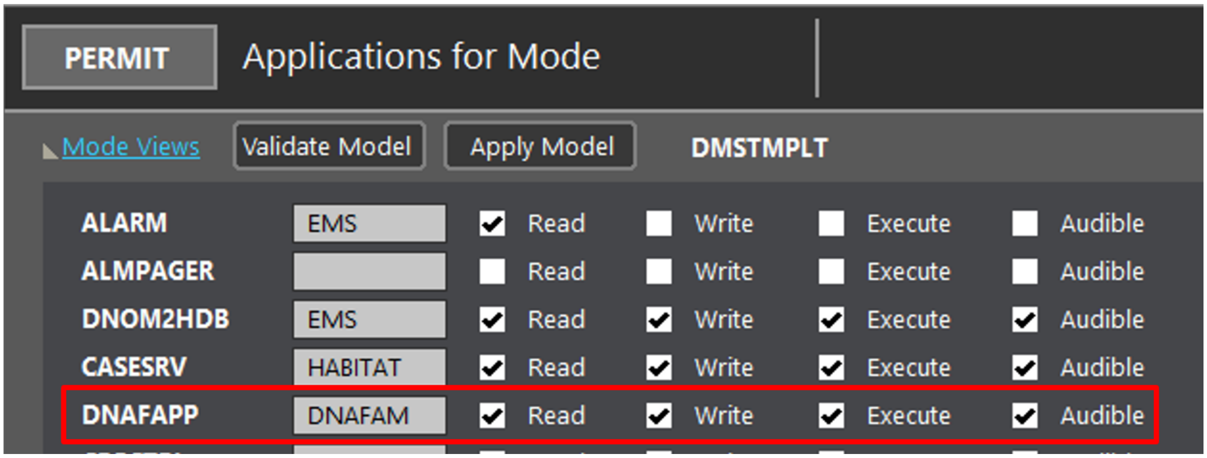


11. Right-click the Navigate field and select Mode Applications.

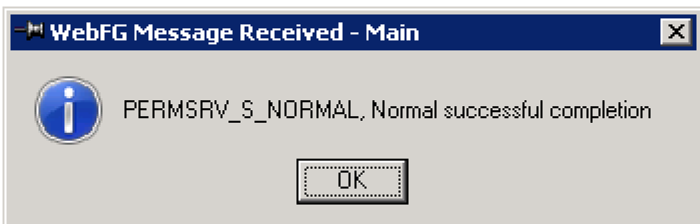


    The Applications for Mode display is opened.

12. Scroll to where the DNAFAPP is added as a PERMIT application.
13. Enter the DNAFAM family name, select the permission boxes, and press Enter.



14. Click Validate Model and check that the model validated successfully.



15. Click Apply Model and check that the model was applied successfully.

3.2. PROCMAN

DNOM2HDB is declared as a PROCMAN task. It can be automatically started after the main PROCMAN tasks are started, or it can be manually started (in this case, do not select the Init State check box for the RUN_DNOM2HDB flag).

3.2.1. PROCMAN Model Changes

This section describes adding the DNOM2HDB task to the PROCMAN model. For more information about configuring the Habitat PROCMAN model, refer the *PROCMAN User Guide*.

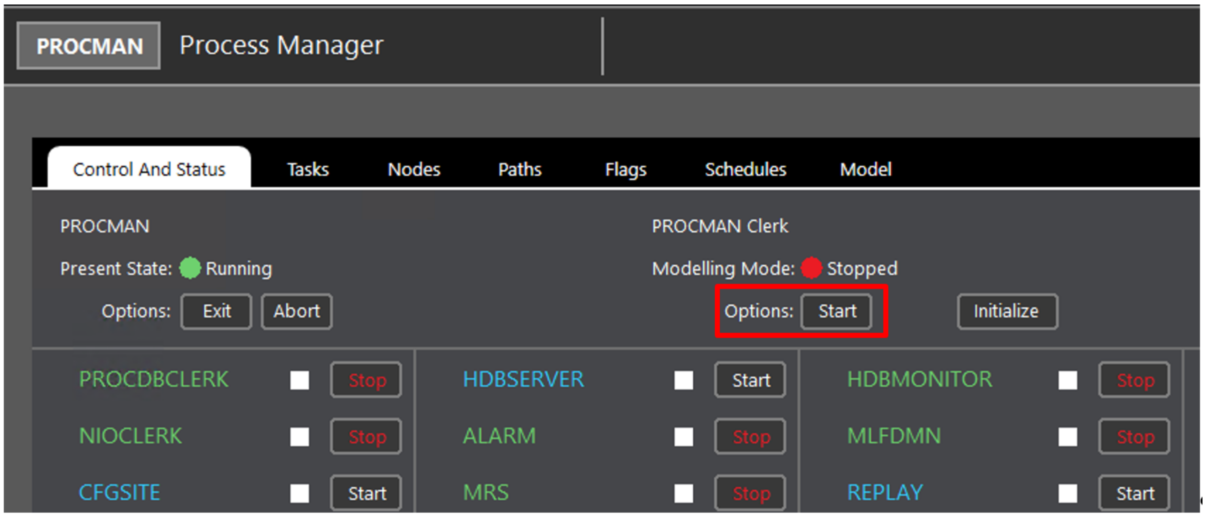
> **Important**
>
> The DNOM2HDB PROCMAN task already exists in the default model and it does not need to be manually added.

To manually add the DNOM2HDB task to the PROCMAN model:

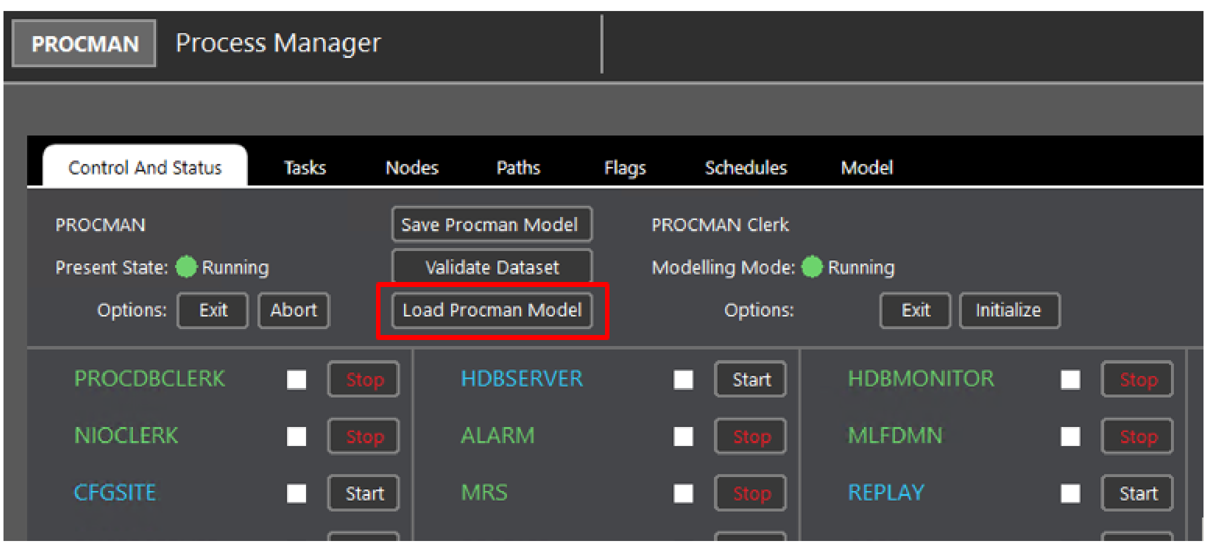
1. Stop Habitat.
2. Start Habitat without loading ADMS-specific configurations by using the command:

```
tview r procman procman habitat
```

3. Start the Browser client via the default Browser shortcut without loading any ADMS-specific configurations.
4. Log in as EMS (without password).
5. Go to the Process Manager display (HABITAT Applications menu > Process Manager > Process Manager Master).
6. Start the PROCMAN Clerk modeling mode.



7. On the Control and Status tab, click Load Procman Model.



8. On the Model Tab, add the DNOM2HDB task, flag, flag conditions, node, and path as shown in DNOM2HDB Task Model image.
9. On the Control and Status tab, select Validate Dataset to verify the model file.

10. Click Save Procman Model to save the model file.
11. Exit the PROCMAN Clerk.
12. Stop Habitat.
13. Stop the Browser client.
14. Start Habitat with ADMS-specific PROCMAN tasks.
15. Start the Browser client with ADMS-specific configurations tasks.

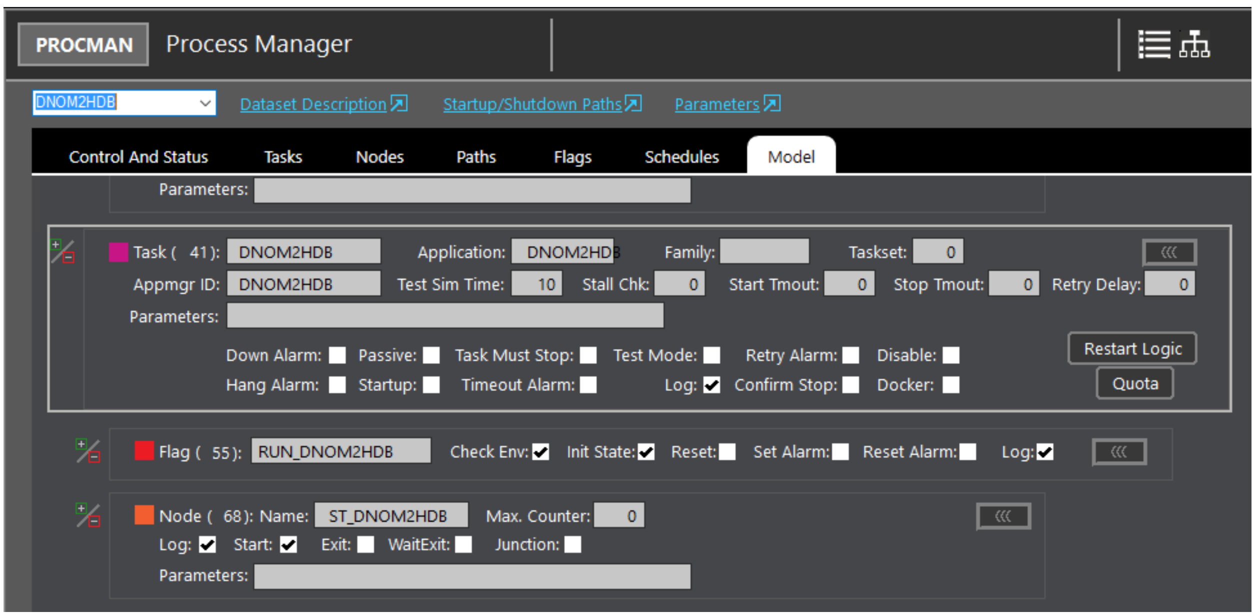


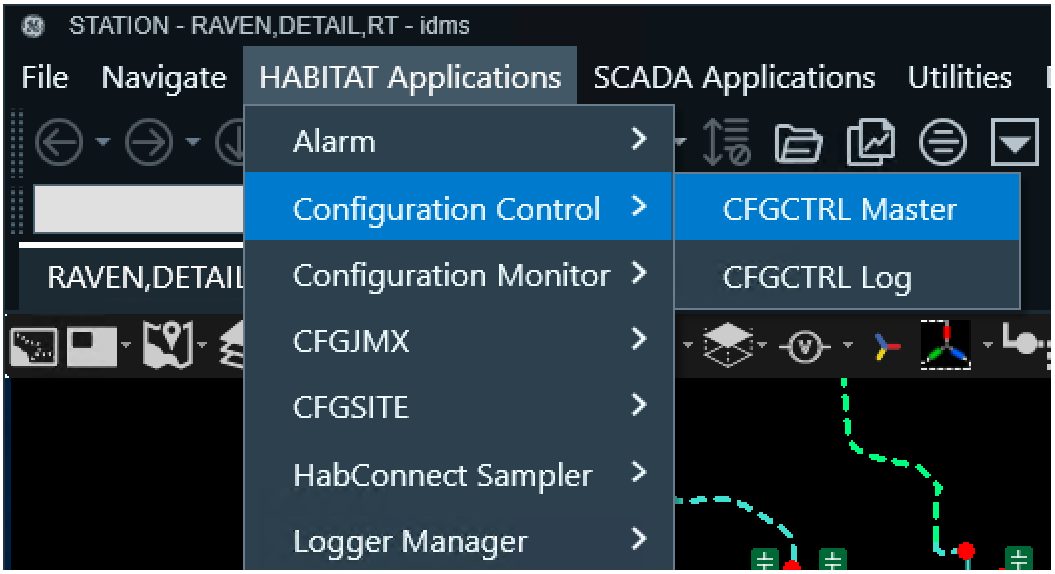
Figure 1. DNOM2HDB Task Model

3.3. Configuration Control

DNOM2HDB must be defined in the Configuration Control model. This configuration is not needed in a simulator environment.

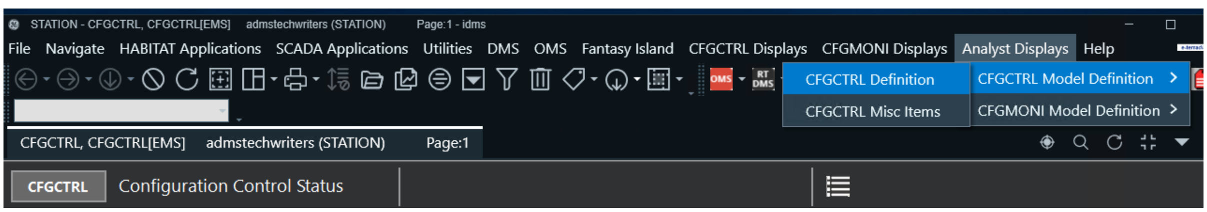
To define DNOM2HDB in the Configuration Control model:

1. Navigate to HABITAT Applications menu > Configuration Control > CFGCTRL Master.



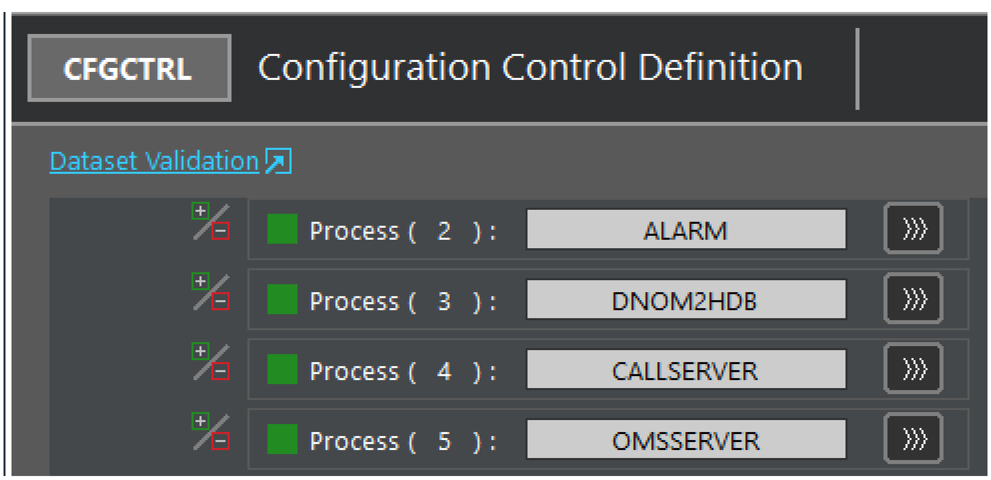
The Analyst Displays menu appears.

2. Navigate to Analyst Displays menu > CFGCTRL Model Definition > Control > CFGCTRL Definition.

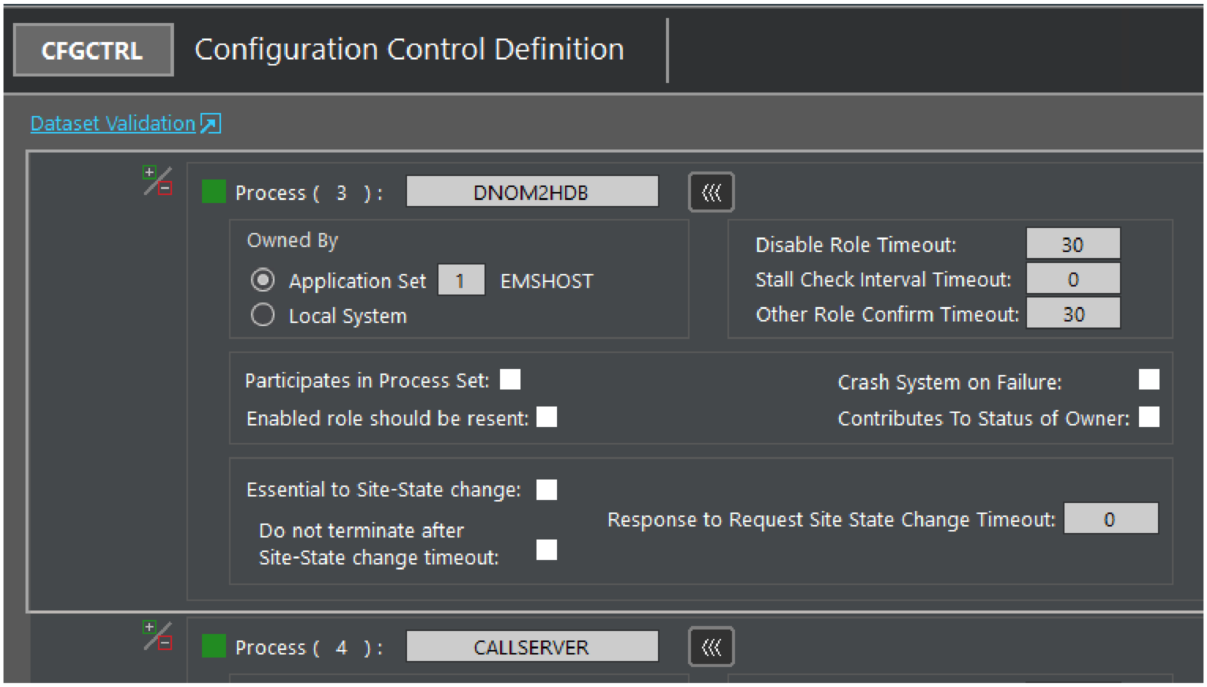


The Configuration Control Definition display opens.

3. In the Configuration Control Definition display, add a new process for DNOM2HDB.

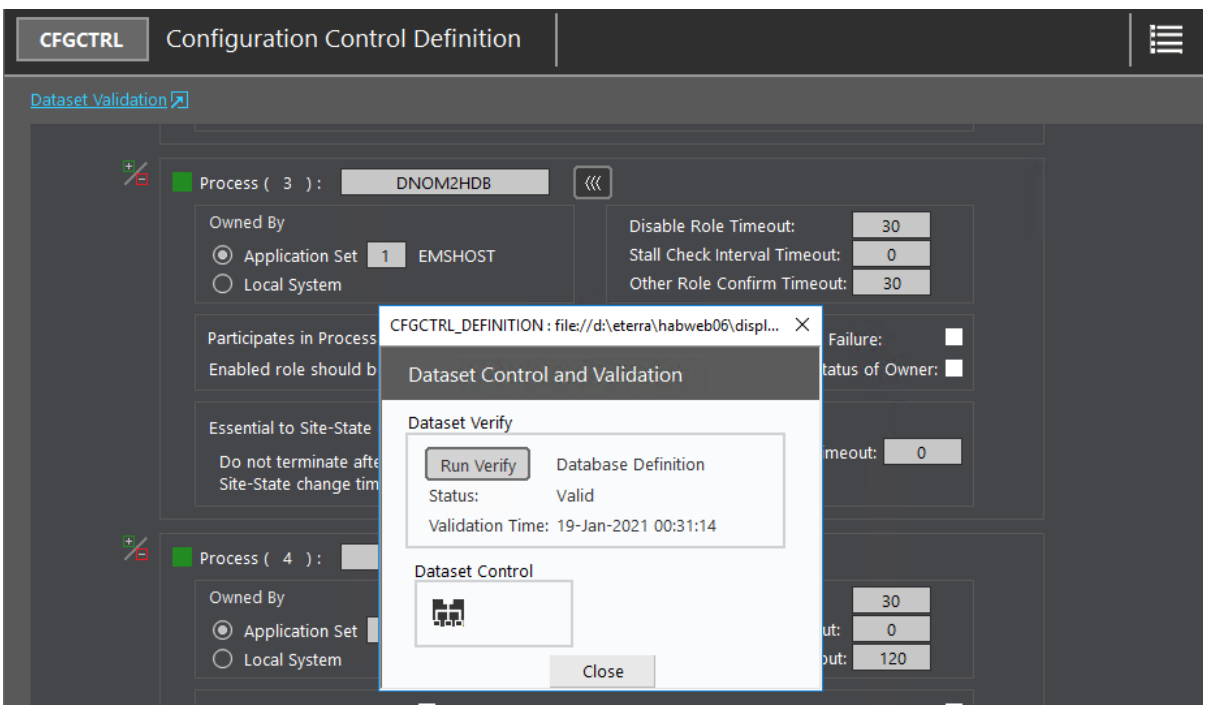


4. Click the expand button (>>>) for the DNOM2HDB process, and define the detail information for this process.

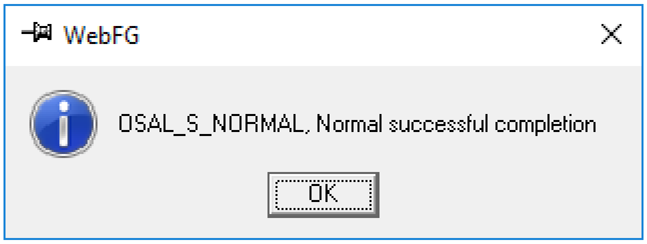


5. Navigate to HABITAT Applications menu > Process Manager > Process Manager Master.
6. Stop the CFGCTRL task.
7. Return to the Configuration Control Definition display with the DNOM2HDB process details.

   The Dataset Control and Validation dialog box opens.



8. Click the Run Verify button to run validation.



9. Start the CFGCTRL task again.

3.4. Displays

Make sure that displays are stored in the following folders:

- Displays from DNOM2HDB are stored in %HABITAT_WEBROOT%\displays\dnom2hdb.
- DNAF displays (DNAF_LVMSTEP, DNAF_FISRSTEP, and DNAF_FISRPLAN) are stored in %HABITAT_WEBROOT%\displays\dnafapp.

3.5. Menus

Add the DNOM2HDB menu items to the HABITAT menu.

1. Open the `webfg_menus.txt` file from: %HABITAT_WEBROOT%\config\menus.
2. Go to the section that configures the HABITAT Applications menu. This menu is intended to provide high-level access to each of the HABITAT applications.

```
menu="MENU2",label="HABITAT Applications"
```

3. Search for the last menu parameter. For example:

```
parent="MENU2_21",label="License Metering Log",command="display/app=lmeter lmeter_log
```

4. Add the menu for the DNOM2HDB items by adding the following information below the last HABITAT menu item. For example:

```
menu="MENU2_22",parent="MENU2",label="Dnom2Hdb",application="DNOM2HDB"
parent="MENU2_22",label="Queries",command="display/app=DNOM2HDB/fam=%disfam% query_summary"
parent="MENU2_22",label="HDB Tables",command="display/app=DNOM2HDB/fam=%disfam% table_summary"
parent="MENU2_22",label="Parameters",command="display/app=DNOM2HDB/fam=%disfam% config"
parent="MENU2_22",label="Stations",command="display/app=DNOM2HDB/fam=%disfam% station_summary"
parent="MENU2_22",label="Log",command="display/app=DNOM2HDB/fam=%disfam% log"
parent="MENU2_22",label="Debug",command="display/app=DNOM2HDB/fam=%disfam% debug"
```

webfg_menus.txt

```
    parent="MENU2_19",label="Data Entry Message Log",command="display/app=rfgutils rfgdice_log"
  menu="MENU2_20",parent="MENU2",label="ET Navigate",application="etnav"
    parent="MENU2_20",label="ET Navigate Log",command="display/app=etnav etnav_log"
  menu="MENU2_21",parent="MENU2",label="License Metering",application="lmeter"
    parent="MENU2_21",label="License Metering Master",command="display/app=lmeter lmeter_master"
    parent="MENU2_21",label="License Metering Log",command="display/app=lmeter lmeter_log"
  menu="MENU2_22",parent="MENU2",label="Dnom2Hdb",application="DNOM2HDB"
    parent="MENU2_22",label="Queries",command="display/app=DNOM2HDB/fam=%disfam% query_summary"
    parent="MENU2_22",label="HDB Tables",command="display/app=DNOM2HDB/fam=%disfam% table_summary"
    parent="MENU2_22",label="Parameters",command="display/app=DNOM2HDB/fam=%disfam% config"
    parent="MENU2_22",label="Stations",command="display/app=DNOM2HDB/fam=%disfam% station_summary"
    parent="MENU2_22",label="Log",command="display/app=DNOM2HDB/fam=%disfam% log"
    parent="MENU2_22",label="Debug",command="display/app=DNOM2HDB/fam=%disfam% debug"
```

5. Save your changes.
6. Restart the Browser client.

   The DNOM2HDB menu is now added to the HABITAT Applications menu.

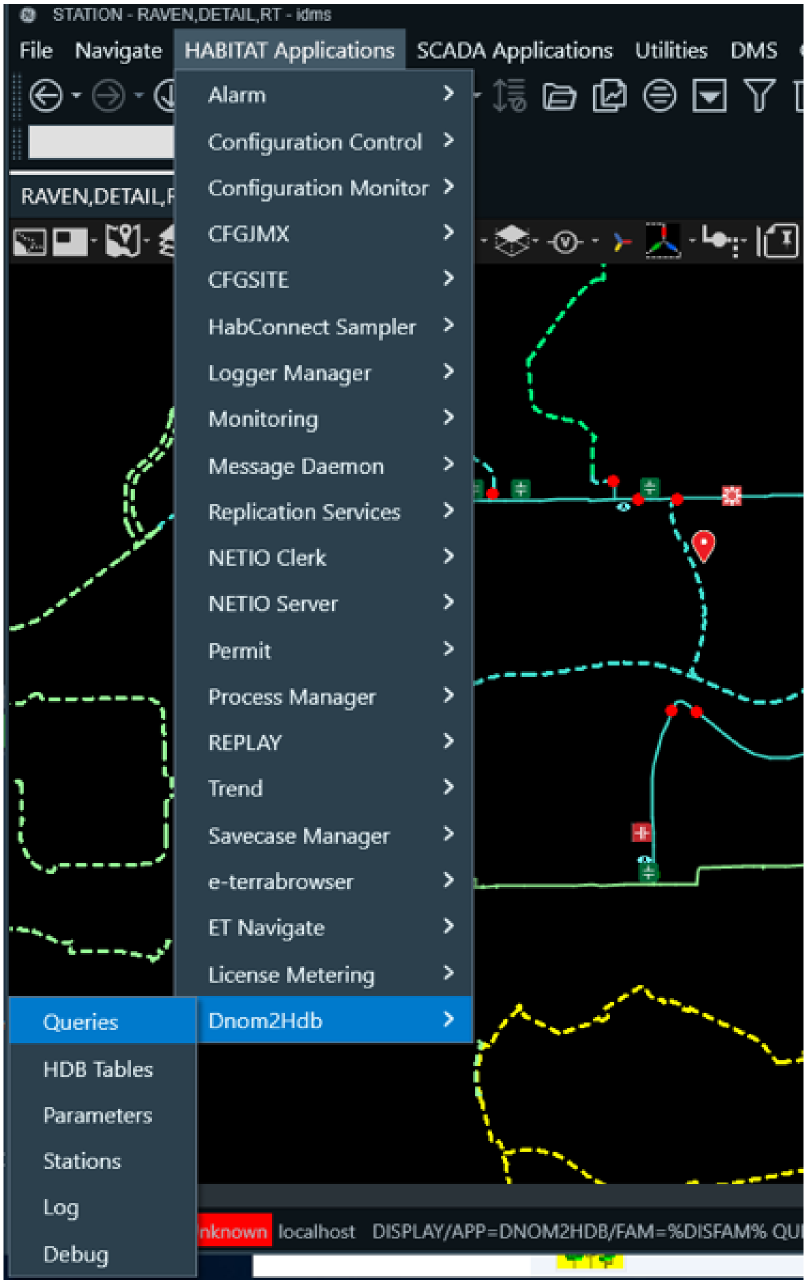


Figure 2. DNOM2HDB Menu

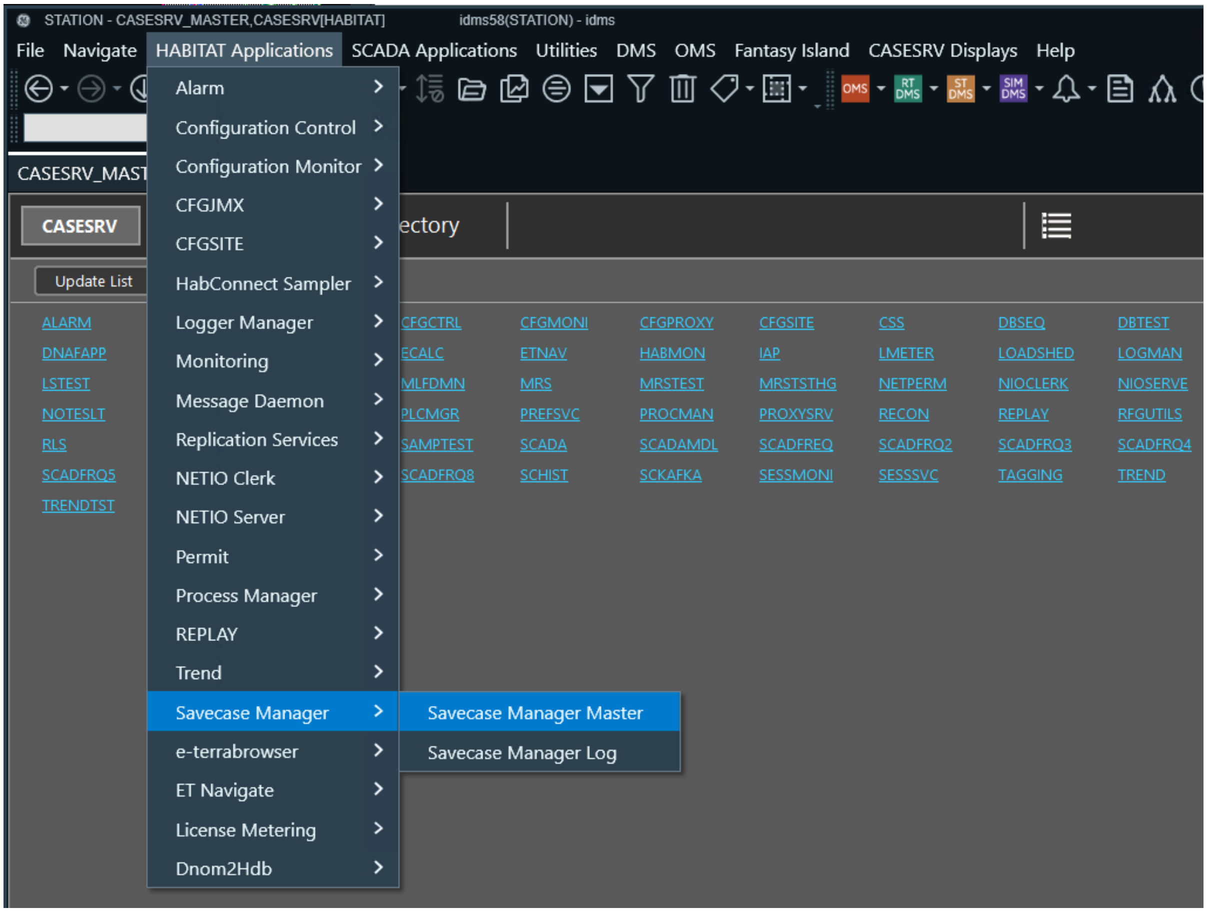
3.6. Savecases

DNOM2HDB has an initial savecase to set up the default values.

To retrieve the initial savecase:

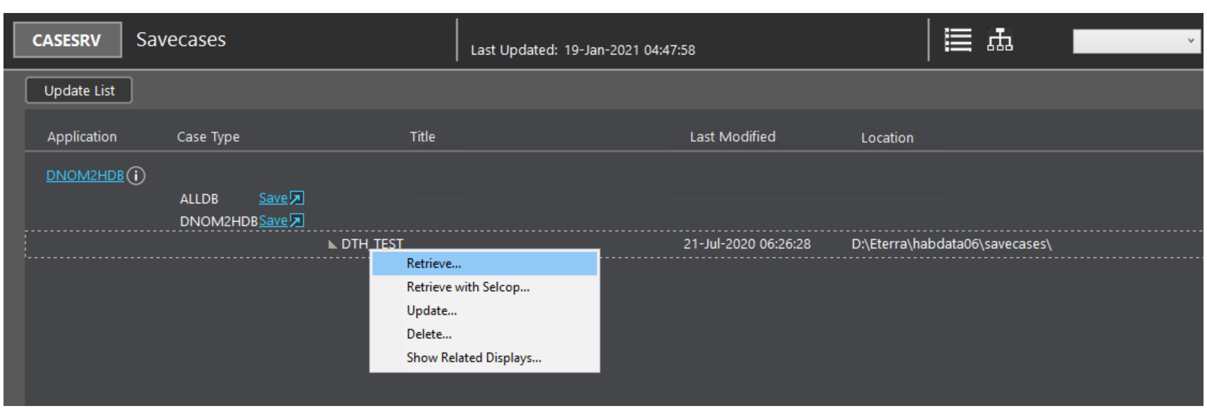
1. Navigate to HABITAT Applications menu > Process Manager > Process Manager Master.
2. Stop the DNOM2HDB task by clicking the Stop button.

3. Navigate to HABITAT Applications menu > Savecase Manager > Savecase Manager Master.

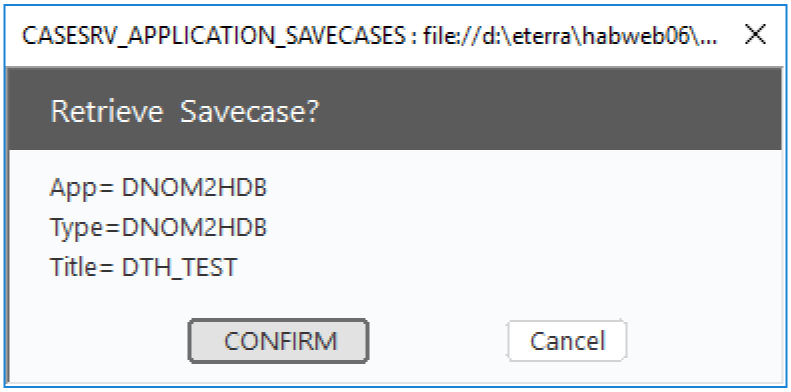


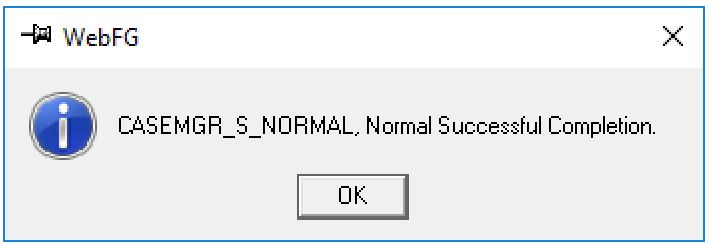
This opens the Savecase Application Directory display.

4. Select the DNOM2HDB task.
5. Right-click on the title and select Retrieve.



6. Click Confirm to retrieve the savecase.





7. Navigate to HABITAT Applications menu > Process Manager > Process Manager Master.
8. Start the DNOM2HDB task.

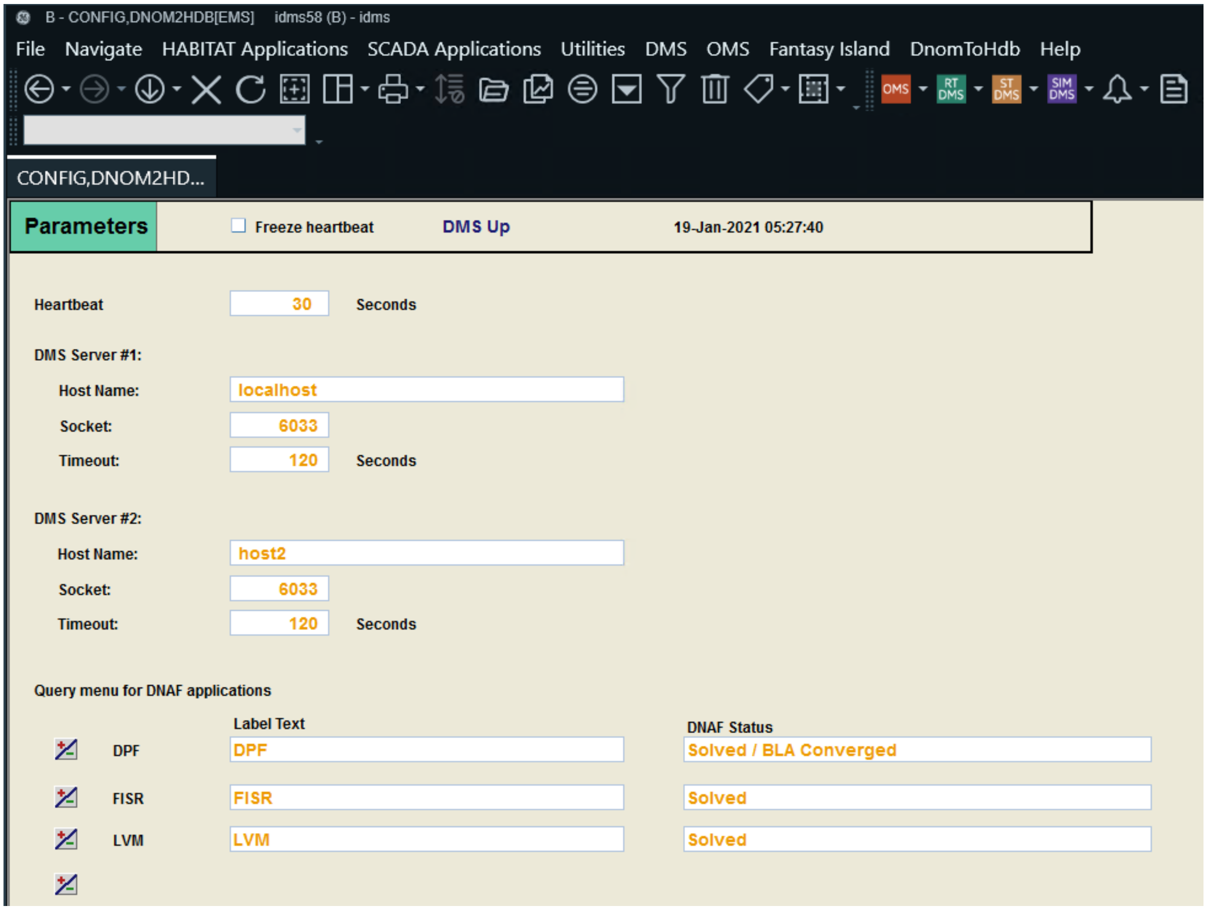
3.7. DMS Service Module

To establish a connection between the DNOM2HDB application and the DMS Service module on the DNAF server:

1. Open the DNAF server configuration file `NetServerDnafConfig.txt` (stored in %IDMS_ROOT%\config\server) in a text editor.
2. Find the "Service module listening socket and socket parameters" section, and uncomment the following parameters:

```
SERVICE_PORT  6033
SERVICE_SOCKET_OUTPUTBUF_MAX_SIZE  6000000
SERVICE_SOCKET_WOULDBLOCK_RETRY  2
SERVICE_SOCKET_MAX_CLIENT_MES_SIZE  640000
```

3. Make sure that SERVICE_PORT is assigned to the port number available on your server.
4. Restart the DNAF server.
5. Open the DNOM2HDB parameters display (HABITAT menu > DNOM2HDB > Parameters).
6. Make sure the Host Name and Socket port of your DMS Service module on the DNAF servers match the DNOM2HDB parameter settings.



Note

The DNOM2HDB process must be started to set the parameters.

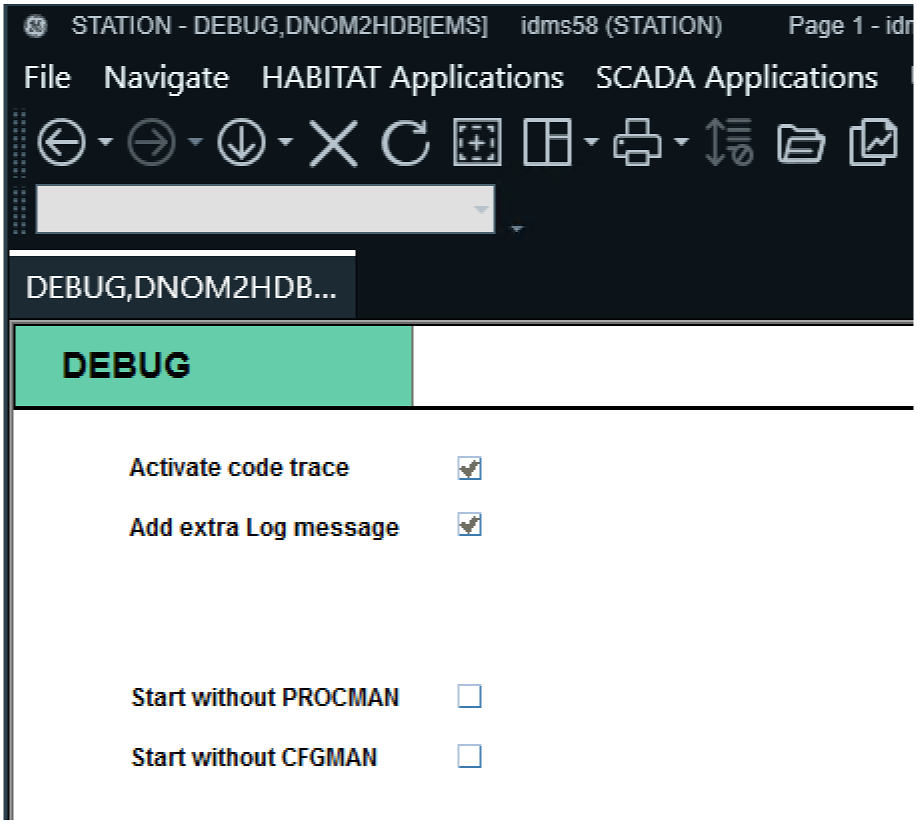
3.8. Troubleshooting

3.8.1. Enabling Detailed Logging

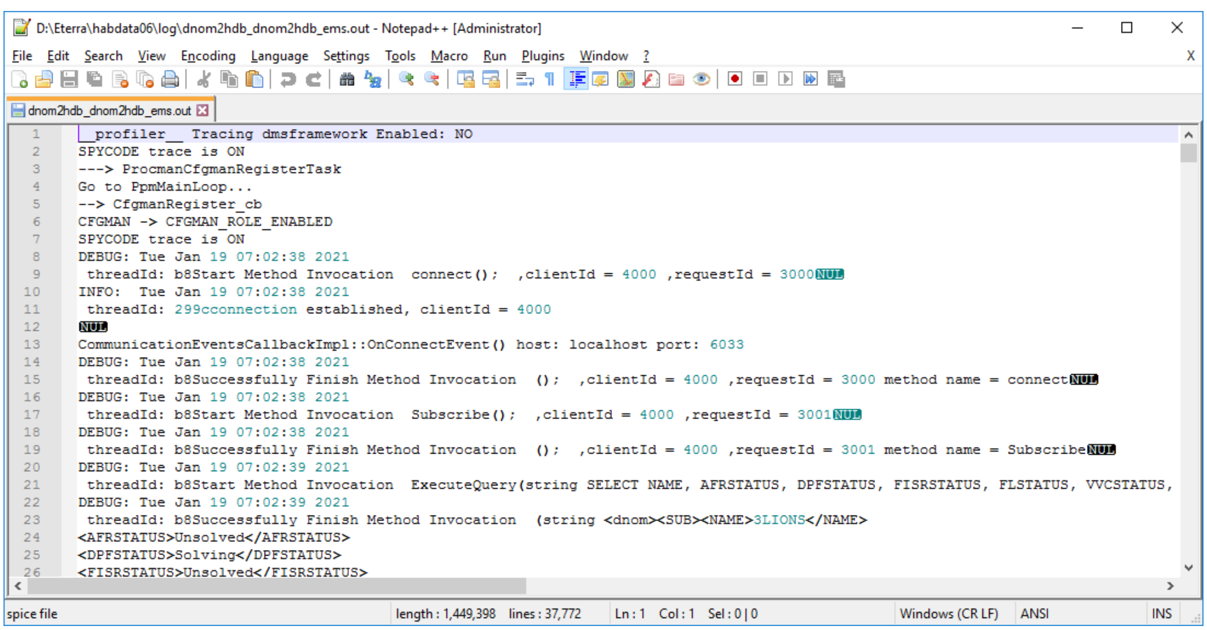
Some extra tracing can be logged in the Log display or to an ASCII output file by selecting the Activate Code Trace option from the Debug display or by defining the SAF_CODE_DEBUG environment variable.

To enable detailed logging from the DNOM2HDB Debug display:

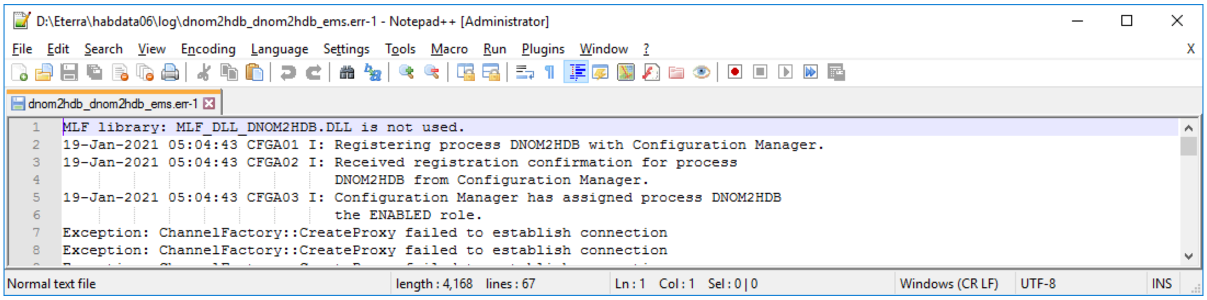
1. Open the DNOM2HDB Debug display.



2. Select the Activate Code Trace and Add Extra Log Message options.
3. Navigate to the %HABITAT_LOGDIR% folder (D:\Eterra\habdata06\log).
4. Open a `dnom2hdb.out` file. This file contains errors and detailed responses for queries that are performed by the DNOM2HDB application.



5. Open a `dnom2hdb.err` file. This file contains application Configuration Manager and Process Manager statuses and errors.

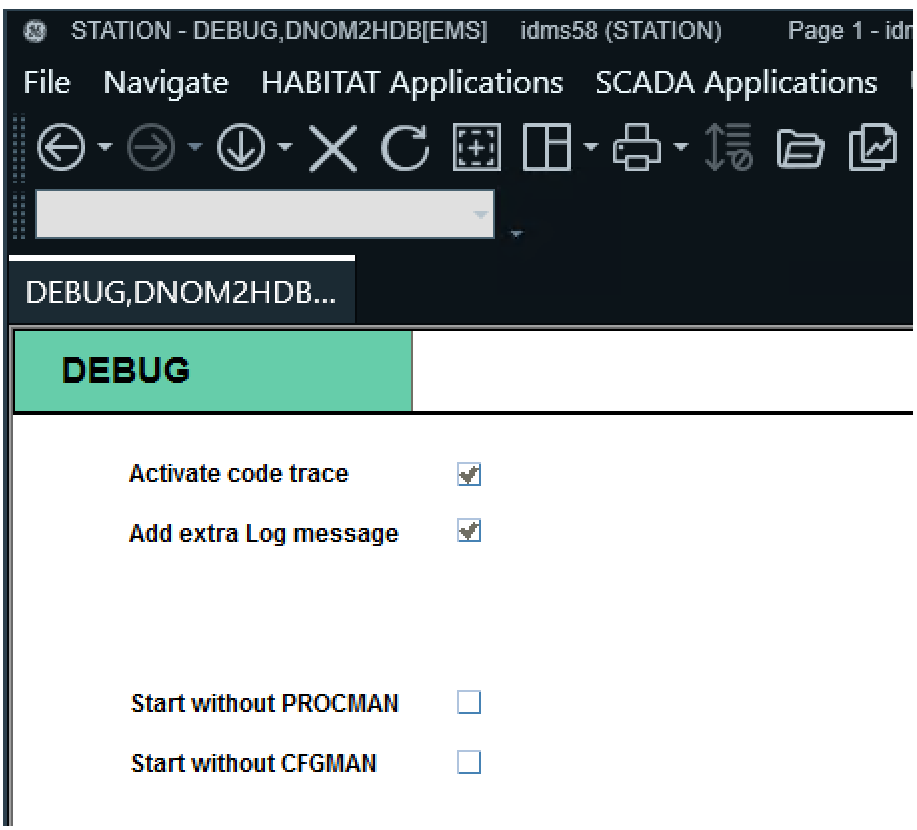


3.8.2. Running DNOM2HDB without PROCMAN

For quick installations, it is possible to run DNOM2HDB even if models of PROCMAN and CFGMAN are not updated.

To run DNOM2HDB without updating the PROCMAN and CFGMAN models:

1. Open the DNOM2HDB Debug display.



2. Select the Start without PROCMAN and Start without CFGMAN options.